

Woods Hole Oceanographic Institution
Department of Physical Oceanography
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Momme Claus Hell

Education

2015–2020	Ph.D. in Physical Oceanography , Scripps Institution of Oceanography, UCSD.
2013–2015	MSc. in Atmospheric and Climate Science , ETH Zürich.
2009–2012	BSc. in Physics of the Earth System , GEOMAR, University of Kiel.

Appointments

OCTOBER 2024	Assistant Scientist (tenure-track) - Woods Hole Oceanographic Institution Department of Physical Oceanography
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Research Experience and Expeditions

2023-2024	Postdoctoral Fellow - National Center for Atmospheric Research (NCAR) Fellow in the Advanced Study Program (2 years) Jointly between Climate & Global Dynamics Lab and Computational & Information Systems Lab Developing a fast wave model for coupled climate models
2022-2023	Visiting Scholar - Scripps Institution of Oceanography (SIO) In the Air-Sea Interaction Laboratory to work on S-MODE data Host: Luc Lenain
2021-2023	Postdoctoral Research Associate - Brown University Parameterization for wave attenuation in sea ice as part of the Scale-Aware Sea Ice Project (SASIP). Advisors: Baylor Fox-Kemper and Chris Horvat
2020-2021	Postdoctoral Researcher - SIO Zonal-mean flows and the occurrence of blocking flow structures in idealized models. Advisor: Nick Lutsko
2015–2020	Graduate Student Researcher - SIO Dissertation title: <i>Atmosphere-Ocean Momentum Exchange by Extra-Tropical Storms</i> . Advisors: Sarah Gille, Bruce Cornuelle, and Arthur Miller
2019	Visiting Graduate Research Student - Ifremer, Brest WaveWatch III school and development of a Lagrangian framework for swell generation. Funding: Julia Brown research fund, Hosts: Bertrand Chaperon and Alex Ayet
2016	Ross Ice Shelf Vibration Project - Expedition to Antarctica Maintaining seismic stations on the Ross Ice Shelf to facilitate continuous wave observations. PI: Peter Bromirski

2013 - 2015	Research Student and Master Student Researcher - ETH Zürich Thesis title: <i>High Arctic-Stratosphere-Feedback on Sub-Seasonal Scale</i> . Master student and student Assistant. Advisors: Tapio Schneider and Noel Keenlyside (UiB).
2013	Research Student Intern - Geophysical Institute, University of Bergen Studying feedback between Arctic sea ice and the Stratospheric Vortex. Advisors: Noel Keenlyside and Martin King.
2012	Undergraduate Student Researcher - GEOMAR Kiel Bachelor Thesis: <i>Influence of the Atlantic Sea Surface Temperature on rainfall in the Sahel and India in Kiel Climate Model</i> . Advisors: Mojib Latif and Claus Böning

Fellowships and Awards

2023 - 2024	Advanced Study Program Postdoctoral Fellowship at NCAR Jointly between CISL and the CGD unit for developing and implementing an efficient surface-wave model for climate models.
2023	NASA Openscapes Champions Program Professional development program for research teams using data from NASA Distributed Active Archive Centers (DAACs)
2019	Julia Brown Prize and ISBlue Mobility Prize For establishing collaborative research with Ifremer Brest
2015 - 2016	UC Regents Fellow – Graduate Student Fellow from the Regents of the University of California
2015	Thesis with Distinction – Awarded by ETH Zürich for excellent thesis work
2010 - 2015	Fellow of the German National Academic Foundation - Studienstiftung des Deutsches Volkes Awarded by the German Nation Academic foundation based on students' excellence

Summer Schools and Workshops

JUNE 2025	The Physics of Changing Polar Climate - Kavli Institute for Theoretical Physics, University of California Santa Barbara, CA - Visting Fellow
JUNE 2018	Rossbypalooza - University of Chicago, Chicago Understanding climate through simple models
SEPT. 2017	Advanced Climate Dynamics Course - Rondane National Park, Norway On dynamics of the seasonal cycle
AUGUST 2017	Les Houches School of Physics - Les Houches, France On fundamental aspects of turbulent flows in climate dynamics

Teaching and Mentoring

2025-PRESENT	Ryan Shijié Dù - PostDoc on NASA SWOT analysis Co-advising with Bia Villas Boas (School of Mines) and Luc Lenain (SIO)
SPRING 2025	12.870 Air-Sea Interaction - MIT/WHOI Joint Program co-teaching with Tom Farrar
2024-PRESENT	Phd-Committee – Tom Protin (PhD Student, Ifremer, Brest, Advisors: Bertrand Chapron, Valentin Resseguier) External committee member and collaborator for wave-current model development
FALL 2022	Sheridan Teaching Seminar at Brown University – to improve inclusive teaching through a reflective exploration of the teachers' own practices and beliefs about teaching.
2021-2022	Mentor – Ding Ding Wei (Computer Science, Brown University) Analysing ICESat-2 data and other Sea Ice products.
SPRING 2020	Guest Lecturer – SIOC 217C <i>Atmospheric and Climate Sciences III: Held and Hou, 1980</i> , UC San Diego (remote).
SUMMER 2019	Mentor – for summer intern Laure Baratgin (École normale supérieure, Paris, now graduate student at LMD and CIRED, Paris) Triangulations with swell observations.
SUMMER 2019	SURF-Mentor – for Amir Rashidi (Computer Science, UCSD) Analysing infra-gravity waves in seismic data.
WINTER 2019	Guest Lecturer – SIO 292 Master of Advanced Studies in Climate Science & Policy Orientation: <i>Ocean Surface Waves in Climate</i> , UC San Diego.
SPRING 2017	Teaching Assistant – SIOC 20 <i>The Atmosphere</i> , UC San Diego.
SUMMER 2013	Co-Instructor – <i>The complexity of the climate system</i> Summer school for climate change and sustainability, Youth education for society and science in cooperation with the National German Academic Foundation, Papenburg, Germany.
2013	Teaching Assistant – " <i>Wetterbesprechung</i> " (<i>Applied Weather Prediction</i>) Undergrad course at GEOMAR, Kiel, Germany.
2012	Teaching Assistant – <i>Climate Physics</i> Undergrad course at GEOMAR, Kiel, Germany.

Funding

Awarded Grants

NASA ROSES-2022 - Studies with ICESat-2, PI. **Momme Hell** and Chris Horvat, *Observing the marginal ice zone with ICESat-2: Improving estimates of sea ice freeboard and storm impacts*, 2023 - 2026, \$440,000

NASA ROSES-2023 - Surface Water and Ocean Topography (SWOT) Science Team, Co-PI. Bia Villas Bôas, **Momme Hell** and Luc Lenain, *Understanding sea state gradients in SWOT observations: opportunities and challenges*, 2024 - 2028, \$983,000 (\$329k for Co-PI)

WHOI Independent Research & Development (IR&D) - 2025, PI: **Momme Hell**, *Towards Stochastic air-sea Fluxes for the Submeso- to Mesoscale*, 2025, \$67,533

Service and Engagement

2024-PRESENT | **US CLIVAR Working Group** - Small scale processes in the upper ocean and their interaction with the Earth's climate system – Member

2024-PRESENT | **US CLIVAR Panel** - Phenomena, Observations, and Synthesis – Member

2017-PRESENT | **Ad-hoc referee for scientific journals**
Reviewer for Journal of Advances in Modeling Earth Systems, Nature Communications, the Journal of Atmospheric Sciences, the Journal of Physical Oceanography, the Journal of Climate, the Journal of Geophysical Research - Oceans, The Cryosphere, and Geophysical Journal International,

2024 | **NASA Science Panel** – Reviewer

2023 | **Climatematch Academy** – Curriculum co-lead for the reanalysis day
Climatematch Academy is a wide-reaching, inclusive, and approachable program aimed to introduce computational methods and create a globally diverse climate sciences community (academy.climatematch.io)

2021-2022 | **Diversity & Inclusion Action Committee** – Participant
Department of Earth, Environmental, and Planetary Sciences, Brown University

2020 | **Session convener** – Convener of “*Mechanisms of Low-Frequency Ocean-Atmosphere Variability and Implications for Earth's Energy Budget*” at the American Geophysical Union Fall Meeting 2018, Washington, DC.

2019 | **Student committee member** – Member of the student committee for the *hydrology and climate faculty search* at Scripps Institution of Oceanography at UC San Diego.

2019 - 2020 | **Scripps Graduate Student Council**
Representative for the Physical Oceanography curriculum group at SIO

2018 - 2019 | **Scripps Climate Student Journal Club**
Hosting weekly journal club series with a quarterly external keynote speaker to facilitate discussions among students and faculty

2008 - 2015 | **Students Helping Life Association**
Member of the board and campaign management
www.schueler-helfen-leben.de

Publications

In Preparation

Hell, M. and Laurence Armi. Atmospheric zonal flows from the perspective of Rossby-beta hydraulics. *in preparation*

Accepted

T. Protin, V. Resseguier, **Hell, M.**, B. Chapron, and R. Fablet. Sequential importance resampling of particle-in-cell for swell dynamics. *accepted in Stochastic Transport in Upper Ocean Dynamics IV*, 2025

Peer-Reviewed

Hell, M., Baylor Fox-Kemper, and Bertrand Chapron. A particle-in-cell wave model for efficient sea-state estimates in earth system models—picles. *Journal of Advances in Modeling Earth Systems*, 2025. doi: doi.org/10.1029/2025MS005221

Hell, M. and C. Horvat. A method for constructing directional surface wave spectra from ICESat-2 altimetry. *The Cryosphere*, 18(1):341–361, January 2024. doi: 10.5194/tc-18-341-2024

Tim Woollings, Li Camille, M Drouard, Etienne Dunn-Sigouin, K Elimestekawy, Brian **Hell, M.** and Hoskins, Cheikh Mbengue, M Patterson, and Thomas Spengler. The role of Rossby waves in polar weather and climate. *EGU Weather and Climate Dynamics*, 2022

Nicholas J. Lutsko and **Hell, M.** Moisture and the Persistence of Annular Modes. *J. Atmospheric Sci.*, 78:3951–3964, December 2021. doi: 10.1175/JAS-D-21-0055.1

Hell, M., Alex Ayet, and Bertrand Chapron. Swell Generation Under Extra-Tropical Storms. *J. Geophys. Res. Oceans*, 126, 2021a. doi: 10.1029/2021JC017637

Hell, M., Bruce D. Cornuelle, Sarah T. Gille, and Nicholas J. Lutsko. Time-Varying Empirical Probability Densities of Southern Ocean Surface Winds: Linking the Leading Mode to SAM and Quantifying Wind Product Differences. *J. Clim.*, 34:5497–5522, July 2021b. doi: 10.1175/JCLI-D-20-0629.1

Hell, M., Tapio Schneider, and Camille Li. Atmospheric Circulation Response to Short-Term Arctic Warming in an Idealized Model. *J. Atmos. Sci.*, 77:531–549, February 2020b. doi: 10.1175/JAS-D-19-0133.1

Hell, M., Sarah T. Gille, Bruce D. Cornuelle, Arthur J. Miller, Peter D. Bromirski, and Alex D. Crawford. Estimating Southern Ocean Storm Positions With Seismic Observations. *J. Geophys. Res. Oceans*, 125, 2020a. doi: 10.1029/2019JC015898

Ana B. Villas Bôas and others including **M. Hell**. Integrated Observations of Global Surface Winds, Currents, and Waves: Requirements and Challenges for the Next Decade. *Front. Mar. Sci.*, 6, 2019. doi: 10.3389/fmars.2019.00425

Hell, M., Bruce D. Cornuelle, Sarah T. Gille, Arthur J. Miller, and Peter D. Bromirski. Identifying Ocean Swell Generation Events from Ross Ice Shelf Seismic Data. *J. Atmos. Oceanic Technol.*, 36:2171–2189, October 2019. doi: 10.1175/JTECH-D-19-0093.1

Martin P. King, **Hell, M.**, and Noel Keenlyside. Investigation of the atmospheric mechanisms related to the autumn sea ice and winter circulation link in the Northern Hemisphere. *Clim Dyn*, pages 1–11, May 2015. doi: 10.1007/s00382-015-2639-5

White Papers

Louise Nuijens and others including **Hell, M.** The air-sea interaction (asi) submesoscale: Physics and impact. *White-paper from the Lorentz-Center Workshop in Leiden, the Netherlands, Sept 2023*, 2024. doi: doi.org/10.5065/78ac-qd31

William J. Merryfield and others including **Hell, M.** Current and Emerging Developments in Subseasonal to Decadal Prediction. *Bull. Am. Meteorol. Soc.*, 101:E869–E896, June 2020. doi: 10.1175/BAMS-D-19-0037.1

J. Cohen and others including **M. Hell**. Arctic change and possible influence on mid-latitude climate and weather. *US CLIVAR Rep*, March 2018. doi: 10.5065/D6TH8KGW

Data and Software

Momme Claus Hell. Particle-in-cell for efficient swell - PiCLES. <https://10.5281/zenodo.13799253>, September 2024

Momme C. Hell. Icesat-2waves: Icesat-2 track analysis to identify and filter ocean surfaces waves. <https://github.com/brown-ccv/icesat2-tracks>, 2024, (under development)

Momme C. Hell. Source code for: “Identifying ocean swell generation events from ross ice shelf seismic data”. <https://doi.org/10.5281/zenodo.5778295>, 2021b

Momme C. Hell. Source code for: “Swell generation under extra-tropical storms”. <https://doi.org/10.5281/zenodo.5201953>, 2021c

Momme C. Hell. Angular momentum budget in spherical pressure coordinates from reanalysis datasets. <https://doi.org/10.5281/zenodo.5808433>, 2021a

Selected Invited Talks

Momme C. Hell, Baylor Fox-Kemper, Brandon Reichl, Jessica Meixner, and Paul Hall. Unified air-sea coupling strategies in eddy-resolving CESM - waves, ice, and everything all at once. Workshop on Synthesizing Research On Ocean Surface Waves In The New Arctic, ICERM Workshop, Providence, RI, USA, 2025-06-30 to 2025-07-02

Momme C. Hell, Baylor Fox-Kemper, and Bertrand Chapron. Particle-in-cell wave model for efficient sea state-dependent parametrization in coupled models. OASIS - Air-Sea Flux from Space Webinar Series, Online, 2025-06-10

Momme C. Hell, Baylor Fox-Kemper, and Bertrand Chapron. Particle-in-cell wave model for efficient sea state-dependent parametrization in coupled models. Scripps Institution of Oceanography - CASPO Seminar, CA, USA, 2025-06-03

Momme C. Hell, Baylor Fox-Kemper, and Bertrand Chapron. Progress towards a particle-in-cell wave model for efficient sea state-dependent parametrization in coupled models. NSF National Center for Atmospheric Research - CGD Seminar, CO, USA, 2025-05-22

Momme C. Hell and Baylor Fox-Kemper. Bulk fluxes in earth system models and beyond - how converging scales may lead to diverging results (talk). NOAA PSL, Boulder, CO, 2024-03-12

Momme C. Hell and Chris Horvat. Learning surface wave spectra and sea ice structure from icesat-2 altimeter - an path to better wave attenuation parametrization in the marginal ice zone. ICESat-2 Science Team Meeting, La Jolla, CA, USA, 2023-10-03

Momme C. Hell and Baylor Fox-Kemper. Bulk fluxes in earth system models and beyond - how converging scales may lead to diverging results. Workshop at the Lorenz Center: Atmosphere-ocean coupling at (sub)mesoscales, Leiden, Netherlands, 2023-09-26

Momme C. Hell. Southern ocean extreme wind statistics and their effect on resonant swell generation under storms. BCCR seminar, GFI, Bergen, Norway, 2023-06-26

Momme C. Hell. When the ocean is not flat - Rough challenges of waves in sea-ice, models, and air-sea coupling. Woods Hole Oceanographic Institution, PO Seminar, 2023-05-03

Momme C. Hell, Alex Ayet, and B. Chapron. Swell generation under extra-tropical storms. AMS 2023 Session on Extreme Maritime Weather, Denver, 2023-01-10. URL <https://ams.confex.com/ams/103ANNUAL/meetingapp.cgi/Session/61789>

Momme C. Hell. Towards data-driven parameterizations with directional surface wave spectra from icesat-2 altimetry. IMSI Remote Sensing for Climate Analysis - Chicago, US, 2022-12-01. URL <https://www.imsi.institute/videos/directional-surface-wave-spectra-and-sea-ice-structure-from-icesat-2-altimetry/>

Momme C. Hell and Chris Horvat. Directional surface wave spectra and sea ice structure from ICESat-2 altimeter. SASIP Meeting 2022 - Grenoble, France, 2022-06-21

Momme C. Hell. How annular modes are linked surface wave events and wind-driven mixing. Atmosphere Ocean Science Colloquium, NYU, New York, 2022-03-30

Momme C. Hell, Alex Ayet, and B. Chapron. Swell generation under extra-tropical storms. MIT, virtual, 2021-10-06

Conference Presentations

Momme C. Hell, Baylor Fox-Kemper, and Bertrand Chapron. A particle-in-cell wave model for improving air-sea coupling climate and prediction models. CELLO – Climate Exploration in Lively Liaison with the Ocean 2025, Hamburg, Germany, 2025-09-17

Momme C. Hell, Baylor Fox-Kemper, and Bertrand Chapron. Particle-in-cell wave model for efficient sea state-dependent parametrization in coupled models (talk). 5th workshop on waves and wave-coupled processes, ECMWF, Reading, UK, 2024-04a

Momme C. Hell, Baylor Fox-Kemper, and Bertrand Chapron. Particle-in-cell wave model for efficient sea state-dependent parametrization in coupled models (talk). WISE Meeting, Corsica, France, 2024-04b

Momme C. Hell, Baylor Fox-Kemper, and Bertrand Chapron. Particle-in-cell wave model for efficient sea state-dependent parametrization in coupled models (talk). AGU Ocean Science Meeting 2024, New Orleans, LA, USA, 2024-02-20

Baylor Fox-Kemper, **Momme C. Hell**, and Chris Horvat. Energy, pv, cascades: how to do them and why to care (talk). AGU Ocean Science Meeting 2024, New Orleans, LA, USA, 2024-02-22

Momme C. Hell, Baylor Fox-Kemper, and Bertrand Chapron. Particle-in-cell for efficient swell (poster). The 29th WISE meeting, Princeton, NJ, 2023-05-09b

Momme C. Hell, Baylor Fox-Kemper, and Bertrand Chapron. Particle-in-cell for efficient swell (poster). CLIVAR workshop on Mesoscale and Frontal-Scale Air-Sea Interactions Workshop, Boulder, CO, 2023-05-09a

Momme C. Hell. Southern ocean extreme wind statistics and their effect on resonant swell generation under storms (virtual talk). MAXSS 2023 : Marine Atmosphere eXtreme Satellite Synergy workshop, Brest, France, 2023-04-04

Momme C. Hell and Baylor Fox-Kemper. A particle-in-cell wave model for efficient sea-state and swell estimates in coupled models (virtual talk). CESM Ocean Model Winter Working Group meeting - Boulder, CO, USA, 2023-02-10. URL https://www.youtube.com/watch?v=K5hXck5L_9A&t=10419s

Momme C. Hell and Chris Horvat. Directional surface wave spectra and sea ice structure from ICESat-2 (poster). Gordon Research Conferences - Ocean Mixing, Mount Holyoke College, MA, USA, 2022-06-06

Momme C. Hell and Chris Horvat. Constraining frequency dependent wave attenuation in sea ice using ICESat-2 photon heights (talk). Ocean Sciences Meeting, virtual, 2022-02-24

Momme C. Hell and Laurence Armi. Atmospheric zonal flows and blocks from the perspective of Rossby-beta hydraulics (talk). Atmospheric Blocking Virtual Workshop, virtual, 2021-09-27. URL <https://blocking-workshop-2021.wavestoweather.de/program/index.html>

Momme C. Hell, Bruce D. Cornuelle, and Sarah T Gille. Leading modes and biases of Southern Ocean Surface Wind from time-varying Distributions (talk). SOCCOM Annual Meeting, virtual, 2021-06

Momme C. Hell, Bruce D. Cornuelle, S. T. Gille, and Nicholas J. Lutsko. Time-varying empirical probability densities of Southern Ocean surface winds: Leading modes linked to SAM, the annual cycle, and product differences (talk). IOVWST Meeting, virtual, 2021-02-24. URL <https://www.youtube.com/watch?v=qE3398GFsvY>

Natalie M Freeman, Donata Giglio, Sarah T Gille, and **Momme C. Hell**. On the impact of high-frequency wind variability on upper ocean stratification (poster). AGU Fall Meeting (virtual), 2020-12-04. URL https://mdc.coaps.fsu.edu/scatterometry/meeting/docs/2021/FREEMAN_AMS21_POSTER.pdf

Momme C. Hell, Tapio Schneider, and Camille Li. Atmospheric Circulation Response to Short-Term Arctic Warming in an Idealized Model (talk). 22th Atmosphere and Ocean Fluid Dynamics Meeting, Portland, Main, 2019-06

Momme C. Hell, Cornuelle, B. D., S. T. Gille, Arthur J. Miller, Peter D. Bromirski, and Alex D. Crawford. Southern Ocean Storm positions and wave attenuation under sea ice estimated with seismic observations in the Ross Ice Shelf (talk). ACDC 10-Year Anniversary Conference, Rondanau, Norway, 2019-03

Momme C. Hell, Martin P. King, and Noel Keenlyside. Investigation of the atmospheric mechanisms related to the autumn sea ice and winter circulation link in the Northern Hemisphere (talk). EGU Meeting, Vienna, Austria, 2015-05

Computational Skills

OPERATING SYSTEMS	Unix-based operating systems and high-performance computing (HPC) environments (NSF NCAR - Dercho/Casper, NASA- Pleiades, ETH - Euler, Brown U. - OSCAR, and small clusters at SIO and WHOI.
PROGRAMMING LANGUAGES	Python, Julia, Latex, Bash, Shell-Script, Fortran, and MatLab.
TOOLS AND SOFTWARE	Jupyter kernels, workflow management with Makefile, version control systems like git, slurm, cdo, Adobe Creative Suite, typo3, html5.
NUMERICAL MODELING	CESM, The flexible Modeling system (FMS), Finite difference models, WaveWatchIII, and Particle-in-Cell

Languages

GERMAN	Native language
ENGLISH	Full proficiency

STATUS September 2025